An Ancient Ocean of Giant Predators Changes Ideas About Evolution

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Long ago, an ocean covered what is now central Colombia, and giant sea reptiles ruled the waves. Long-necked **plesiosaurs** swam beside sea crocodiles, turtles, sharks and **crustaceans**. Now **fossils** from that lost world are changing how scientists think about evolution. Scientists from McGill University and Colombia studied **fossils** from the Paja Formation, near Villa de Leyva. The rocks, from the Early Cretaceous period, preserve a picture of an ocean community called the Paja Biota. Ancient animal groups from the Jurassic period, already heading toward **extinction**, lived side by side with newer Cretaceous groups just beginning to flourish. The food web was also very intricate, with an unusually large variety of top **predators**. Hans Larsson and Dirley Cortés call this pattern a 'Biotic Inflection,' when older groups nearing **extinction** live alongside newer ones starting to expand. The scientists think a 'perfect storm' made it possible: warm temperatures, high sea levels and nutrient-rich waters. The big idea is that major changes in life on Earth do not always need a mass **extinction**. But more evidence is needed: more **fossils** and better dates.

Word Treasure

plesiosaurs -- Ancient sea animals with long necks that lived in the oceans long ago.
fossils -- The remains of animals or plants that lived long ago, turned to stone.
crustaceans -- Sea animals with hard shells, like crabs and shrimps.
predators -- Animals that hunt and eat other animals.
extinction -- When a whole kind of animal or plant dies out and disappears.

Background you need

Fossils are the remains of animals and plants that lived long ago, turned to stone in rock.

Evolution is the slow change of living things over millions of years.

A food web shows who eats whom, from tiny creatures up to the top predators.

Break it down

WHO: Scientists from McGill University and Colombia, led by Hans Larsson and Dirley Cortés.
WHAT: Fossils from an ancient ocean are changing ideas about evolution.
WHERE: In the Paja Formation, near Villa de Leyva in Colombia.
WHY: Old and new animal groups lived side by side, which scientists had not expected.

Why it matters

If you love dinosaurs or sea creatures, this story shows how fossils can change what scientists think about life long ago.

Test what you remember

1. Where were the fossils found?

- (A) The Paja Formation, near Villa de Leyva in Colombia
- (B) The coast of Australia
- (C) The Amazon rainforest
- (D) A desert in Egypt

2. Which animals ruled the ancient ocean?

- (A) Dinosaurs on land
- (B) Whales and dolphins
- (C) Giant sea reptiles such as plesiosaurs
- (D) Giant penguins

3. What is special about the animals that shared the sea?

- (A) Old groups lived beside newer groups
- (B) Only one species lived there
- (C) All the animals were babies
- (D) No predators lived there

4. What name did the scientists give the pattern?

- (A) A 'Biotic Inflection'
- (B) A 'Perfect Storm'
- (C) A 'Mass Extinction'
- (D) A 'Food Web'

5. What made the 'perfect storm' possible?

- (A) A big earthquake
- (B) Warm temperatures, high sea levels and nutrient-rich waters
- (C) A meteor strike
- (D) Cold temperatures and low sea levels

6. What do scientists say is still needed?

- (A) More fossils and better dates
- (B) A bigger team
- (C) A new museum
- (D) More money for space travel

Answer key (try the quiz first!): 1: A 2: C 3: A 4: A 5: B 6: A

Questions to think about

1. Professor Hans Larsson: He says the team has discovered a new model of how life on Earth changes, a 'Biotic Inflection'.

2. The research team: It stresses that more evidence is needed, with more fossils and better dates.

MY THOUGHTS (at least 20 words):